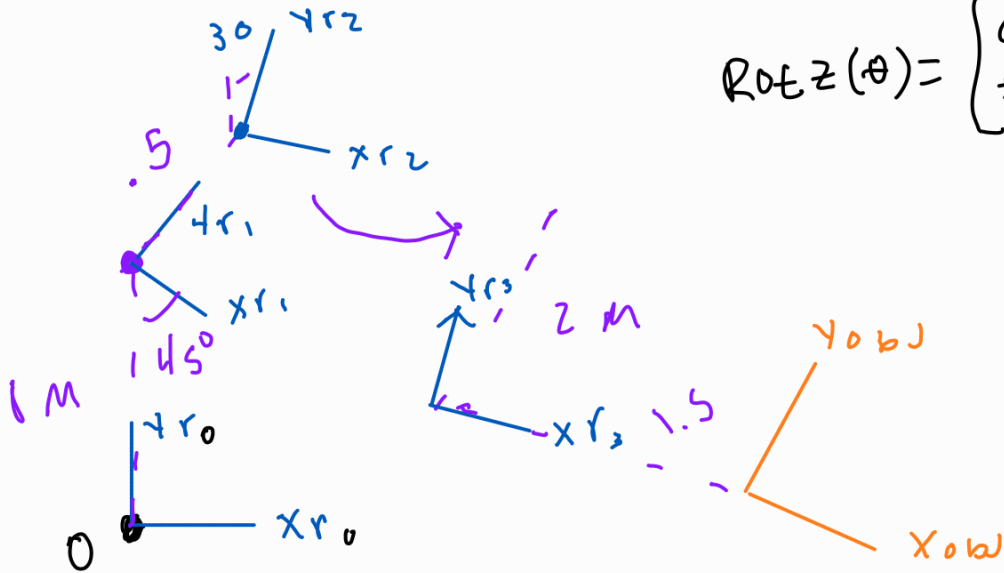


Problem 1

a)



$$\text{Rot}_Z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

b)

$${}^0T_{\text{Final}} = \text{Trans} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{Rot}(-45) \\ \text{Trans} \begin{pmatrix} 0 \\ .5 \end{pmatrix} \text{Rot}(15) \\ \text{Trans} \begin{pmatrix} 0 \\ -2 \end{pmatrix}$$

c)

$${}^0T_{\text{Final}} = {}^0T_1 {}^1T_2 {}^2T_3$$

$$\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & .5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_{\text{Final}} = \begin{bmatrix} .9659 & .2588 & -.1641 \\ -.2588 & .9659 & -.5782 \\ 0 & 0 & 1 \end{bmatrix}$$

$$d) {}^F T_{obj} = \begin{bmatrix} {}^F R_{obj} & {}^F d_{F \rightarrow obj} \\ \vec{0}^T & 1 \end{bmatrix}$$

$$\text{Final } T_{obj} = \begin{bmatrix} 1 & 0 & 1.5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

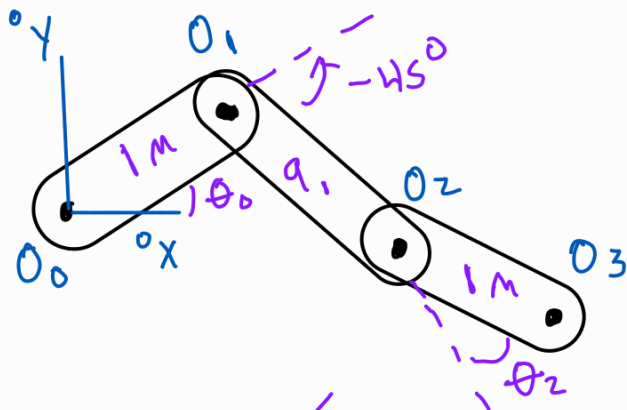
$$e) {}^0 T_{obj} = {}^0 T_{final} \cdot \text{Final } T_{obj}$$

$$= \begin{bmatrix} .9659 & .2588 & 1.2848 \\ -.2588 & .9659 & -.9665 \\ 0 & 0 & 1 \end{bmatrix}$$

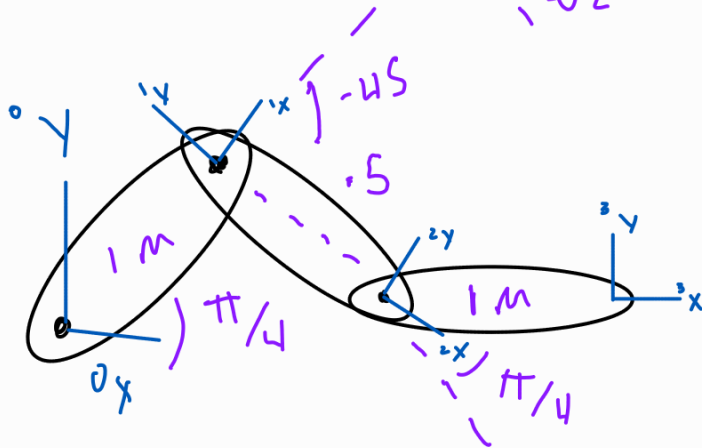
$${}^{obj} T_0 = {}^0 T_{obj}^{-1} = \begin{bmatrix} {}^0 R_{obj}^{-1} & -{}^0 R_{obj}^{-1} {}^0 d_{0 \rightarrow obj} \\ \vec{0} & 1 \end{bmatrix}$$

$${}^{obj} T_0 = \begin{bmatrix} -.9659 & .2588 & -1.4556 \\ -.2588 & -.9659 & .6105 \\ 0 & 0 & 1 \end{bmatrix}$$

Problem 2



a)



$${}^0R_1 = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \therefore$$

b)

$${}^0T_1 = \begin{bmatrix} {}^0R_1 & {}^0d_{01} \\ \underline{\underline{0}} & 1 \end{bmatrix}$$

$${}^0R_1 = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$${}^1d_{01} = 1 \hat{x} \quad {}^0d_{01} = 1 \cos\theta \hat{x} + 1 \sin\theta \hat{y}$$

$$\therefore {}^0T_1 = \begin{bmatrix} \cos\theta_0 & -\sin\theta_0 & \cos\theta_0 \\ \sin\theta_0 & \cos\theta_0 & \sin\theta_0 \\ 0 & 0 & 1 \end{bmatrix}$$

c)

$${}^1T_2 = \begin{bmatrix} {}^1R_2 & {}^1d_{12} \\ \vec{0} & 1 \end{bmatrix}$$

$${}^2R_1 = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 \\ \sin\theta_1 & \cos\theta_1 \end{bmatrix} \therefore {}^1R_2 = \begin{bmatrix} \cos\theta_1 & \sin\theta_1 \\ -\sin\theta_1 & \cos\theta_1 \end{bmatrix}$$

$$\theta_1 = -45^\circ \therefore {}^1R_2 = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

$${}^2d_{12} = a_1 \hat{z} \quad {}^1d_{12} = a_1 \cos\theta_2 \hat{x} - a_1 \sin\theta_2 \hat{y}$$

\$\therefore\$

$${}^1T_2 = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 & a_1 \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 & -a_1 \sqrt{2}/2 \\ 0 & 0 & 1 \end{bmatrix}$$

d)

$${}^2T_3 = \begin{bmatrix} {}^2R_3 & {}^2d_{23} \\ \vec{0} & 1 \end{bmatrix}$$

$${}^2R_3 = \begin{bmatrix} \cos\theta_2 & \sin\theta_2 \\ -\sin\theta_2 & \cos\theta_2 \end{bmatrix} \quad {}^3R_2 = \begin{bmatrix} \cos\theta_2 - \sin\theta_2 \\ \sin\theta_2 & \cos\theta_2 \end{bmatrix}$$

$${}^3d_{23} = 1 \quad {}^3\hat{X} \quad {}^2d_{23} = \cos\theta_2 \quad {}^2\hat{X} + \sin\theta_2 \quad {}^2\hat{Y}$$

$$\therefore {}^2T_3 = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & \cos\theta_2 \\ \sin\theta_2 & \cos\theta_2 & \sin\theta_2 \\ 0 & 0 & 1 \end{bmatrix}$$

E)

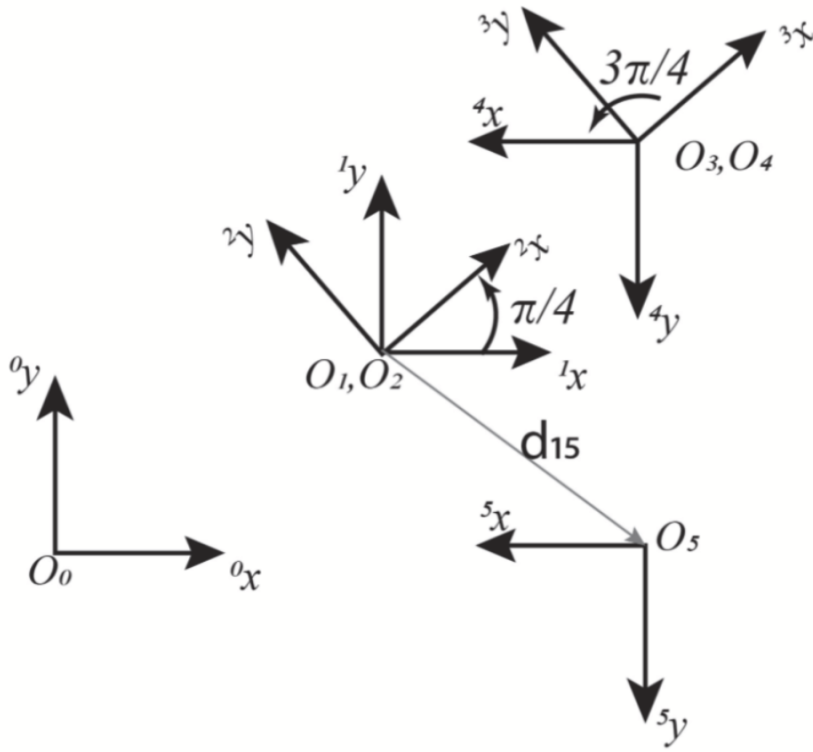
$${}^0T_3 = {}^0T_1 \quad {}^1T_2 \quad {}^2T_3$$

$$\begin{bmatrix} \cos\theta_0 & -\sin\theta_0 & \cos\theta_0 \\ \sin\theta_0 & \sin\theta_0 & \sin\theta_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 & a_1\sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 & -a_1\sqrt{2}/2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & \cos\theta_2 \\ \sin\theta_2 & \cos\theta_2 & \sin\theta_2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{\sqrt{2}}{2} \cos\theta_2 + \frac{\sqrt{2}}{2} \sin\theta_2 & \frac{\sqrt{2}}{2} \cos\theta_2 - \frac{\sqrt{2}}{2} \sin\theta_2 & \frac{\sqrt{2}}{2} \cos\theta_2 + \frac{\sqrt{2}}{2} \sin\theta_2 + a_1 \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \cos\theta_2 + \frac{\sqrt{2}}{2} \sin\theta_2 & \frac{\sqrt{2}}{2} \cos\theta_2 + \frac{\sqrt{2}}{2} \sin\theta_2 & -\frac{\sqrt{2}}{2} \cos\theta_2 + \frac{\sqrt{2}}{2} \sin\theta_2 - a_1 \frac{\sqrt{2}}{2} \\ 0 & 0 & 1 \end{bmatrix}$$

$$OT_3 = \begin{bmatrix} C(\theta_0 - 45^\circ + \theta_2) - 5(\theta_0 - 45^\circ + \theta_2) & C\theta_0 + a_1 C(\theta_0 - 45^\circ) + C(\theta_0 - 45^\circ + \theta_2) \\ C(\theta_0 - 45^\circ + \theta_2) & 5\theta_0 + a_1 5(\theta_0 - 45^\circ) + 5(\theta_0 - 45^\circ + \theta_2) \\ 0 & 1 \end{bmatrix}$$

Problem 3



$${}^0d_{01} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad {}^1d_{12} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad {}^2d_{23} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$${}^3d_{34} \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad {}^4d_{45} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

a) ${}^0T_1 = \begin{bmatrix} {}^0R_1 & d_{01} \\ \vec{0} & 1 \end{bmatrix}$ ${}^0R_1 = \underline{\underline{\hat{t}}}$

$$T_1 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$b) {}^1T_2 = \begin{bmatrix} {}^1R_2 & {}^1d_{12} \\ \vec{0}^T & 1 \end{bmatrix} \quad {}^1R_2 = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\theta = \pi/4$$

$${}^1T_2 = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & 0 \\ \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad {}^1R_2 = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

$$c) {}^2T_3 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\sin, \cos(\pi/4) = \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}$$

$$d) {}^3T_4 = \begin{bmatrix} -\frac{\sqrt{2}}{2} & -\sqrt{2}/2 & 0 \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$e) {}^4T_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

$$F) {}^0T_5 = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4 {}^4T_5$$

$${}^0T_5 = \begin{bmatrix} -1 & 0 & 3.41 \\ 0 & -1 & -.59 \\ 0 & 0 & 1 \end{bmatrix}$$

$$G) \underline{d}_5 = \begin{bmatrix} 3.41 \\ -.59 \end{bmatrix}$$

$$H) \underline{R}_5 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$p = \begin{bmatrix} d_{ij} \\ 1 \end{bmatrix}$$

$$i) \quad {}^s p_{s1} = {}^0 T_s^{-1} \cdot \begin{bmatrix} {}^0 d_{01} \\ 1 \end{bmatrix}$$

$$I) \quad {}^s d_{1s} = -{}^s d_{s1}$$

$$\begin{bmatrix} {}^1 d_{1s} \\ 1 \end{bmatrix} = {}^1 T_s \begin{bmatrix} {}^s d_{1s} \\ 1 \end{bmatrix}$$

$$II) \quad {}^0 T_s^{-1} = \begin{bmatrix} {}^0 R_s^{-1} & -{}^0 R_s^{-1} \cdot \underline{d}_{0s} \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 3.41 \\ 0 & -1 & -1.59 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^s d_{s1} = \begin{bmatrix} 1.41 \\ -1.59 \end{bmatrix}$$

$$\therefore {}^s d_{1s} = \begin{bmatrix} -1.41 \\ 1.59 \end{bmatrix}$$

Problem 4

a)

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\det = 0 \cdot (\text{sum stuff})$$

$$- 1(1 - 0)$$

$$+ 0(\text{sum stuff})$$

$$= -1$$

$\therefore$ not a rotation

b)

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\text{each col norm} = 1$$

$$C_i \cdot C_j = 0$$

$$\det(B) = 1$$

yes rotation

c)

$$\|C_i\|_2 = 1, C_i \cdot C_j = 0, \det(C) = 1$$

yes rotation

d)

$$\|C_i\|_2 \neq 1$$

$\therefore$ not rotation

Problem 3

a)
$$\begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad C_1 \times C_2 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

b)
$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad C_1 \times C_2 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

c)
$$\begin{bmatrix} .3698 & .3352 & .8665 \\ -.9276 & .1858 & .324 \\ -.05 & -.9236 & .3797 \end{bmatrix} \quad C_2 = C_3 \times C_1 = \begin{bmatrix} .3352 \\ .1858 \\ -.9235 \end{bmatrix}$$

d) used python to solve using $\|C\|_2 = 1$
 $C_1 \cdot C_2 = 0$
 $\det = 1$
 properties

$$\begin{bmatrix} -.4049 & .1166 & -.9049 \\ -.2039 & .9550 & .2153 \\ .8893 & .2727 & -.3673 \end{bmatrix}$$